

CS432: Databases

Database Design and the E-R Model

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11

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The Design Process

- Design of the database schema,
- Design of the programs that access and update the data, and
- Design of a security scheme to control access to data

Design Phases

- Understanding user needs. **Prepare specification document.**
- **Prepare conceptual-design** by translating requirements into a conceptual schema of the database. **Entity-relationship (E-R) diagram**
- Designer reviews the schema to ensure it meets **functional requirements.**
- Abstract data model to the implementation of the database:
 - **Logical-design** phase
 - **Physical-design** phase

Avoid two major pitfalls

- Redundancy
- Incompleteness

Any examples from University schema?

The Entity-Relationship Model

The E-R model maps meanings and interactions of real-world enterprises onto a conceptual schema.

- Entity sets
- Relationship sets
- Attributes

A nice diagrammatic representation, **E-R diagrams**.

Entity Sets

Entity

- An entity is a “**thing**” or “**object**” in the real world that is distinguishable from all other objects. For example, instructor in university relation.

Properties/Attributes

- An entity has a set of properties, and the values for some set of properties may uniquely identify an entity. For example, *instructor_id* in instructor relation.

Entity set

- A set of entities of the same type that share the same properties, or attributes. For example, entity set *instructor*.

Entity sets do not need to be disjoint. **Any examples?**

Relationship Sets

A relationship is an association among several entities.

An example

An **advisor** relationship that associates instructor 'Katz' with student 'Shankar'.

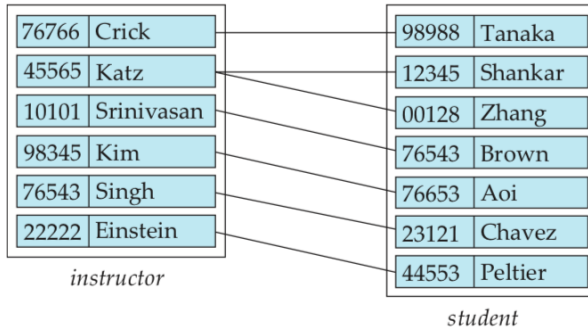
Relationship set

Set of relationships of the same type. If $E_1, E_2, E_3, \dots, E_n$ are entity sets, then a relationship set R is a subset of

$$\{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$$

where $(e_1, e_2, \dots, e_n)$ is a relationship.

Relationship Sets



Relationship Sets

Participation

- The association between entity sets. $E_1, E_2, E_3, \dots, E_n$ **participate** in relationship set R.

Roles

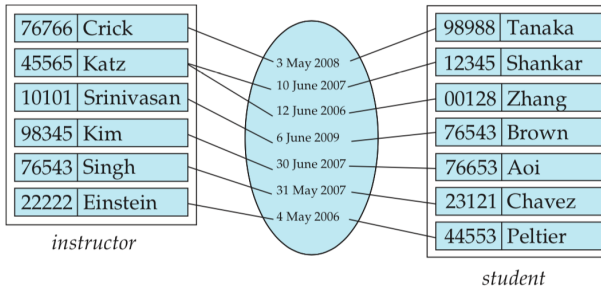
- Each entity plays a distinct *role* in a relationship.

Any examples?

Descriptive attributes

- Attributes of a relationship.

Descriptive attributes



Attributes

For each attribute, there is a set of permitted values, called the **domain**, or **value set**.

Three attribute types:

- Simple and composite attributes
- Single-valued and multivalued attributes
- Derived attribute

Any examples?

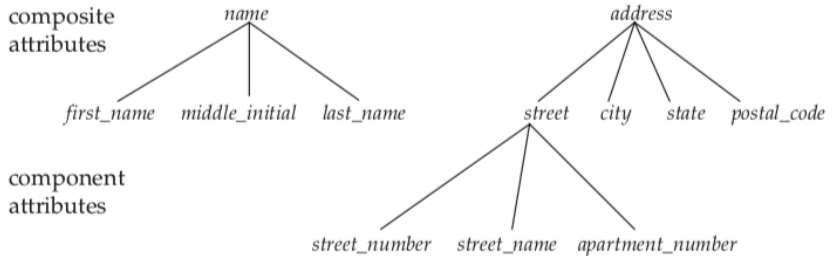
Attributes

For each attribute, there is a set of permitted values, called the **domain**, or **value set**.

Three attribute types:

- Simple and composite attributes (**name in instructor entity set**)
- Single-valued and multivalued attributes (**phone_number in instructor entity set**)
- Derived attribute (**students_advised in instructor entity set**)

Hierarchy in composite attributes



Constraints

Mapping Cardinalities

It expresses the number of entities to which another entity can be associated via a relationship set.

- One-to-one
- One-to-many
- Many-to-one
- Many-to-many

Mapping Cardinalities

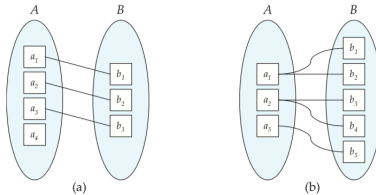


Figura: (a) One-to-one. (b) One-to-many

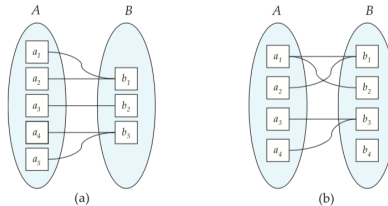


Figura: (a) Many-to-one. (b) Many-to-many.

Participation Constraints

Total

if every entity in E participates in at least one relationship in R .

Partial

If only some entities in E participate in relationships in R .

Keys

A key for an entity is a set of attributes that suffice to distinguish entities from each other.

Relationship set R has no attributes associated with it

$$\text{primary-key}(E_1) \cup \text{primary-key}(E_2) \cup \dots \cup \text{primary-key}(E_n)$$

Relationship set R has $a_1, a_2, \dots, a_n$ attributes associated with it.

$$\text{primary-key}(E_1) \cup \text{primary-key}(E_2) \cup \dots \cup \text{primary-key}(E_n) \cup \{a_1, a_2, \dots, a_m\}$$

A Superkey

$$\text{primary-key}(E_1) \cup \text{primary-key}(E_2) \cup \dots \cup \text{primary-key}(E_n)$$

Removing Redundant Attributes in Entity Sets

Consider the entity sets **instructor** and **department**:

- The entity set **instructor** includes the attributes ID, name, dept_name, and salary, with ID forming the primary key.
- The entity set **department** includes the attributes dept_name, building, and budget, with dept_name forming the primary key.

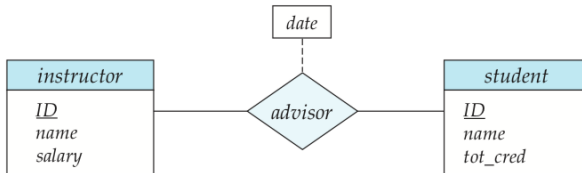
Create a relationship set **inst_dept** relating instructor and department.

dept_name is redundant in instructor.

Entity-Relationship Diagrams

Major components:

- Rectangles divided into two parts
- Diamonds
- Undivided rectangles
- Lines
- Dashed lines
- Double lines
- Double diamonds



Mapping Cardinality



(a)



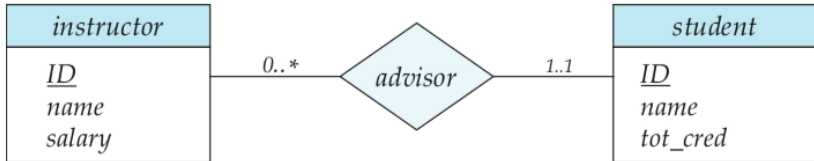
(b)



(c)

Figura: (a) One-to-one. (b) One-to-many. (c) Many-to-many.

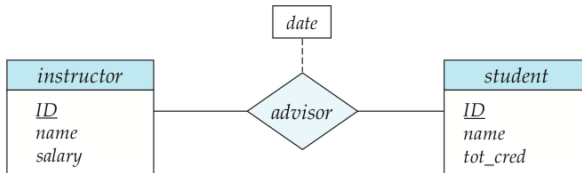
Cardinality Limits



Entity-Relationship Diagrams

Major components:

- Rectangles divided into two parts
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Mapping Cardinality



(a)



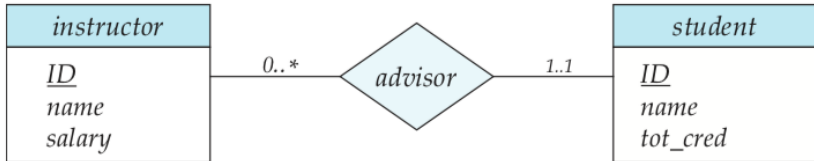
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(c)

Figura: (a) One-to-one. (b) One-to-many. (c) Many-to-many.

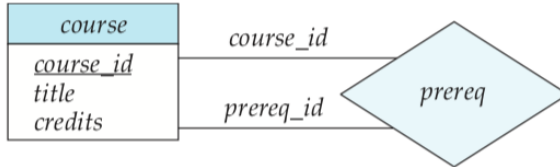
Cardinality Limits



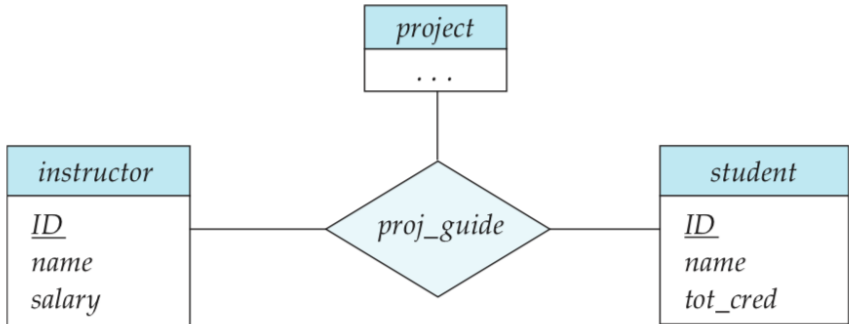
Complex Attributes

<i>instructor</i>
<u><i>ID</i></u>
<i>name</i>
<i>first_name</i>
<i>middle_initial</i>
<i>last_name</i>
<i>address</i>
<i>street</i>
<i>street_number</i>
<i>street_name</i>
<i>apt_number</i>
<i>city</i>
<i>state</i>
<i>zip</i>
{ <i>phone_number</i> }
<i>date_of_birth</i>
<i>age</i> ()

Roles



Nonbinary Relationship Sets



Weak Entity Sets

Section: course_id, semester, year, section_id

Course: course_id, title, credits

- Create a relationship set sec_course
- What information becomes redundant?
- What are the possibilities?



Weak Entity Sets

Section: course_id, semester, year, section_id

Course: course_id, title, credits

- Create a relationship set sec_course
- **What information becomes redundant?**
- What are the possibilities?
 - Get rid of relationship sec_course
 - To not store the attribute course_id in section entity

The second option leads to “section does not have enough attributes to identify a particular section entity uniquely”

Create a special relationship that provides extra information

Weak Entity Sets

- An entity set that does not have sufficient attributes to form a primary key is termed a **weak entity set**.
- It must be associated with another entity set, called the **identifying** or **owner entity set**.
- The relationship associating the weak entity set with the identifying entity set is called the **identifying relationship**.
- **Many-to-one** from the weak entity set to the identifying entity set.
- The participation of the weak entity set in the relationship is **total**.

Weak Entity Sets

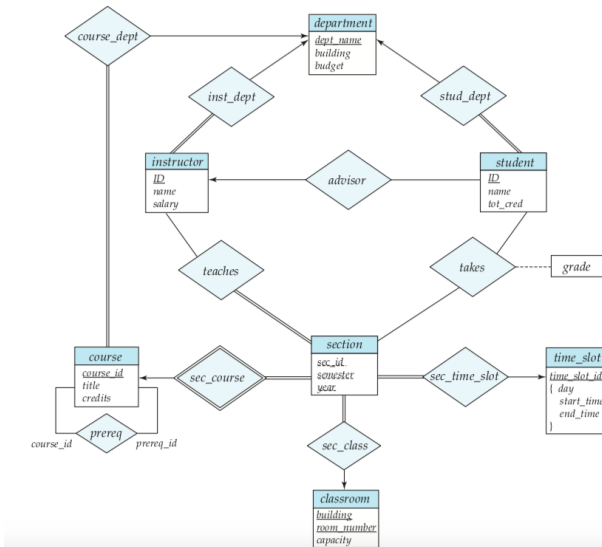
- Weak entity set does not have a primary key.
- How to distinguish among all those entities in the weak entity set that depend on one particular strong entity?
- Partial keys (**Discriminator**) can be helpful.
- section_id, year, and semester
- Primary key of a weak entity set = primary key of the identifying entity set + partial key of the weak entity set.

Weak Entity Sets



- The discriminator of a weak entity is underlined with a dashed line.
- The relationship set is depicted by a double diamond.

E-R diagram for the University Enterprise



Acknowledgments/Contributions

- Some of the images utilized in these slides are subject to copyright - Abraham Silberschatz, Henry Korth, and S. Sudarshan. Database System Concepts. 6th Edition, McGraw-Hill Education
- Contributor 2024-25, 2025-26 - Yogesh K. Meena, IIT Gandhinagar